

active. Now install the piezobuzzer inside the common room of your house. After making proper interconnections, check whether the circuit is working

properly.

Note. To get louder sound, you may replace the piezobuzzer with an AC bell and the 12V DC power supply

with 230V AC mains. You may also connect a manual push-to-on switch between the pole and N/O contacts of the relay for manual operation. ●

MUSICAL WATER SHOWER

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Won't it be nice to have music playing in the background all the while when you take a shower? This simple circuit does the same. It plays different tunes repeatedly for as long as your shower is turned on. The music starts as soon as water comes out of the shower. The music stops when you turn the shower 'off' and water stops coming out of it.

Fig. 1 shows the circuit of the musical water shower. It comprises transistors T1 and T2 which form a complementary amplifier pair, transistor T3 acting as a switch and a 12-tone melo-

dy generator IC M3482. The M3482 is a mask-ROM-programmed IC designed to play melodies according to the programmed data. Its inbuilt preamplifier provides a simple interface to the driver circuit formed by transistors T4 and T5. The IC can be replaced with other UM348xx series, WR630173 or WE4822 melody generator ICs.

The melody section receives power supply through the conduction of transistor T3 when the shower is turned on.

The overhead shower unit is fitted with two insulated copper cables AD and BC as shown in Fig. 2. Insulation of a part of the cable AD (marked A'D') is removed. Cable AD is firmly fixed

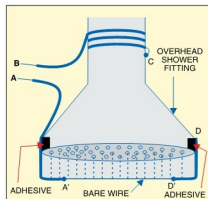


Fig. 2: Sensor arrangement

and routed along the body of shower without any shorting of the bare part with the shower body. The bare part should stretch such that the shower water falls on it when running. One end of copper cable BC is soldered to the body of shower at C. It is wound around the neck to secure it firmly. Thus the cable ends at A and B are available for connection as sensor input to the circuit of Fig. 1. Now when you open the shower tap, water falls on the bare copper wire A'D' which makes electrical contact with the cable BC through the body of the shower which is also in contact with running water. The points A and B of cable on the shower unit are connected as

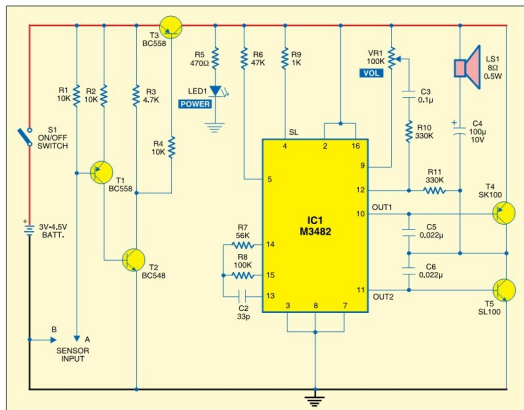


Fig. 1: Circuit for musical water shower